

05 (tidyverse) 基础绘图函数-plot函数

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写在前面：从本节课开始我们将使用 `tidyverse` 包对这个系列中的教学内容进行重置，AI交互与代码编写可以看视频，编程结果在Rproj中，笔记里是对用到的api（函数接口）进行简要介绍，**HAVE FUN!**



本课课纲

- 介绍如何使用 `tidyverse` 画出Base R（对应原版第五节课，基础绘图函数-plot函数）中的图。
- 介绍绘图函数的若干参数。

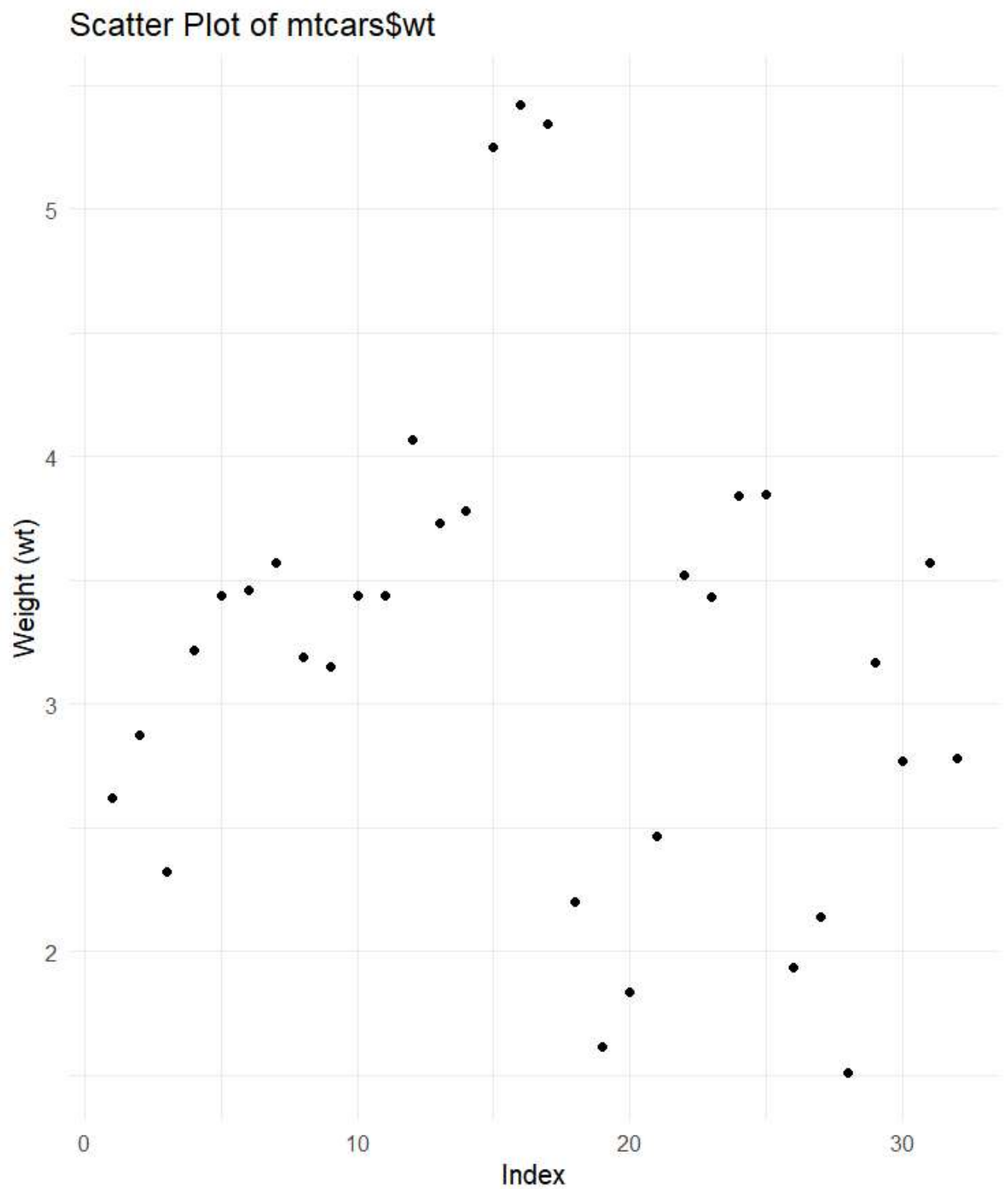
绘制散点图

基础图像

- 绘制单个变量的散点图
 - 代码如下：

```
mtcars %>%  
  mutate(idx = row_number()) %>% # 创建索引列  
  ggplot(aes(x = idx, y = wt)) +  
  geom_point() +  
  labs(  
    x = "Index",  
    y = "Weight (wt)",  
    title = "Scatter Plot of mtcars$wt"  
  ) +  
  theme_minimal()
```

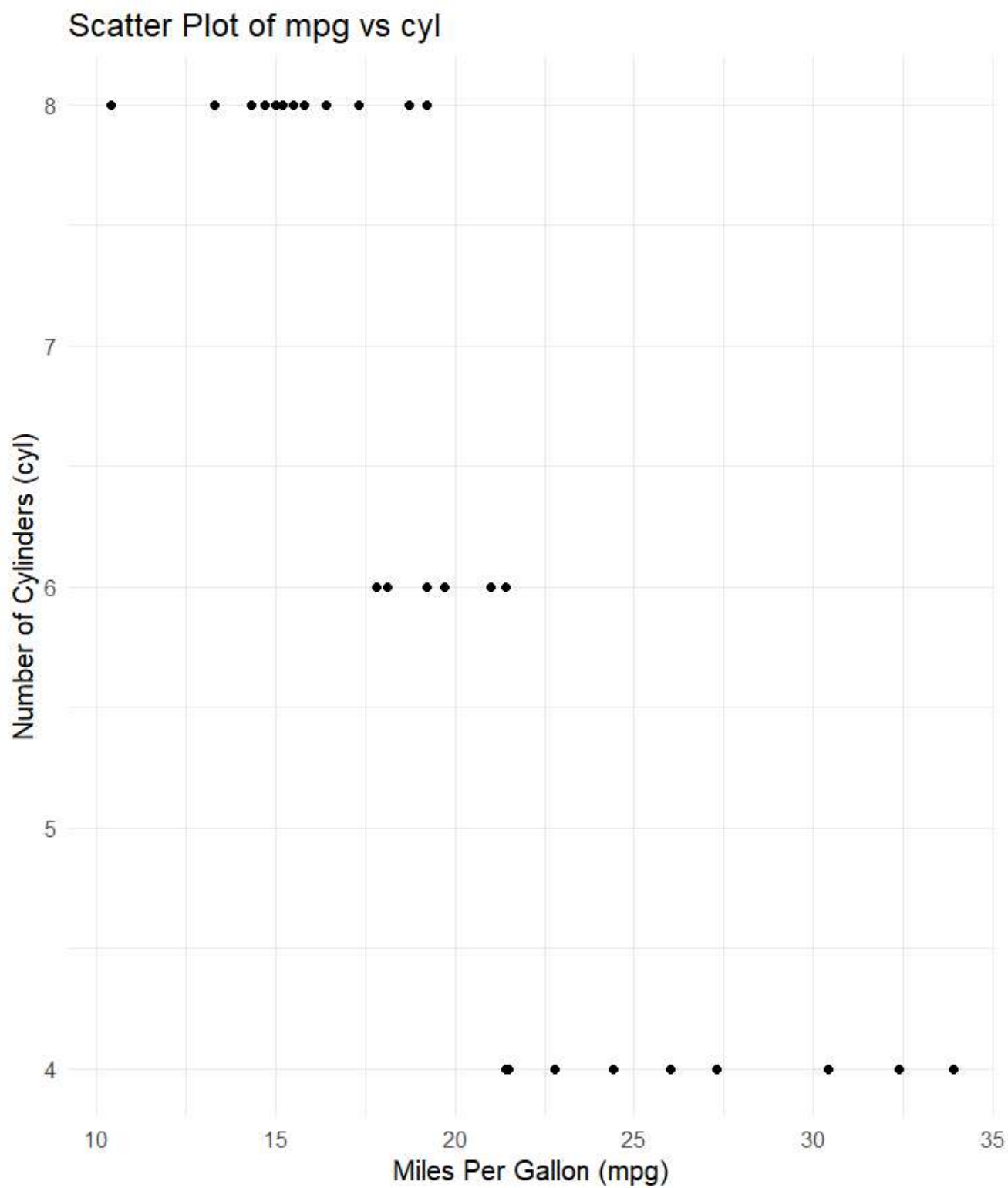
- 绘制结果如下：



- 绘制两个变量的散点图
 - 代码如下：

```
ggplot(mtcars, aes(x = mpg, y = cyl)) +  
  geom_point() +  
  labs(  
    x = "Miles Per Gallon (mpg)",  
    y = "Number of Cylinders (cyl)",  
    title = "Scatter Plot of mpg vs cyl"  
  ) +  
  theme_minimal()
```

- 绘制结果如下：

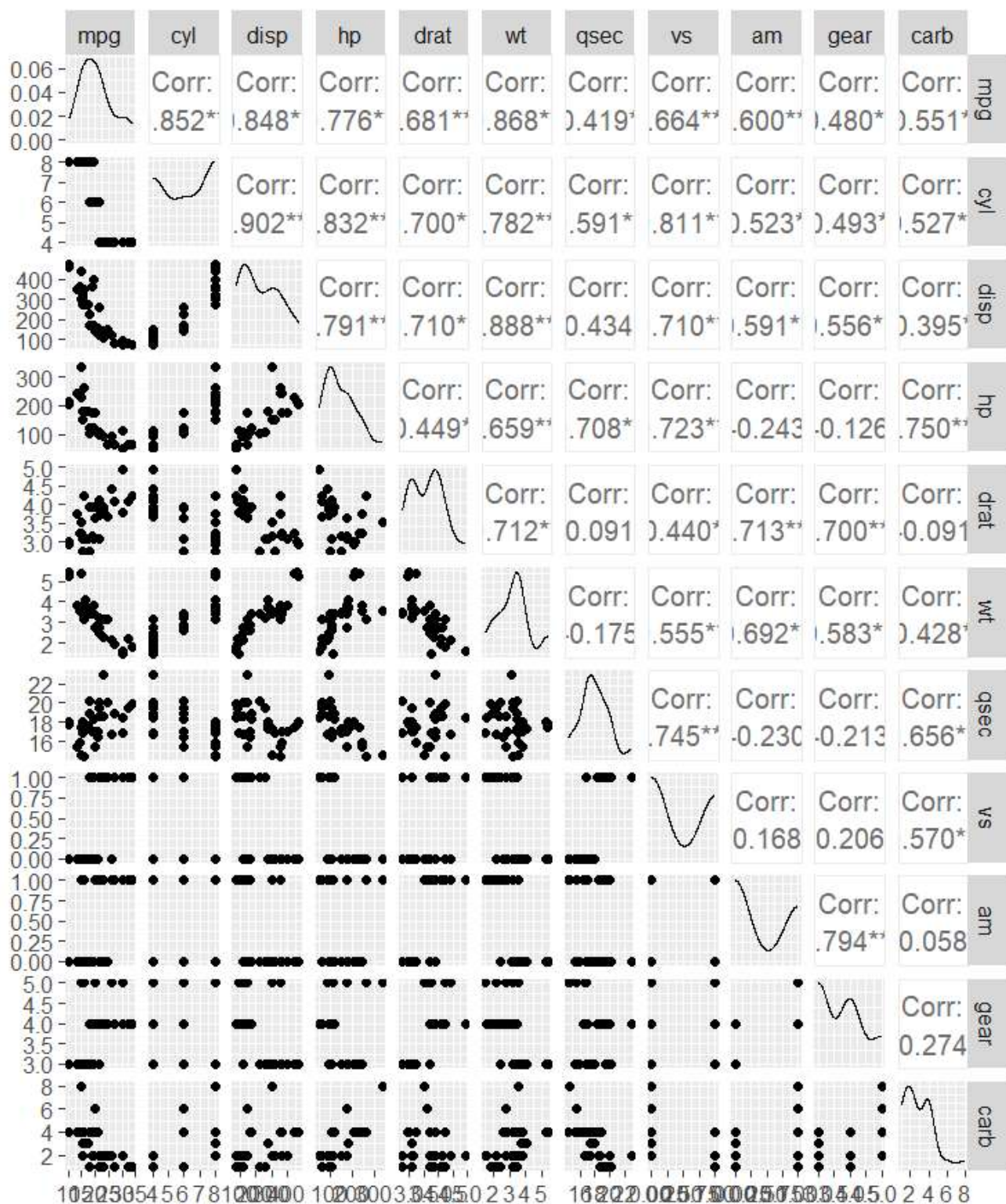


- 绘制数据框两两变量之间的散点图

- 代码如下：

```
library(GGally)
ggpairs(mtcars)
```

- 绘制结果如下：

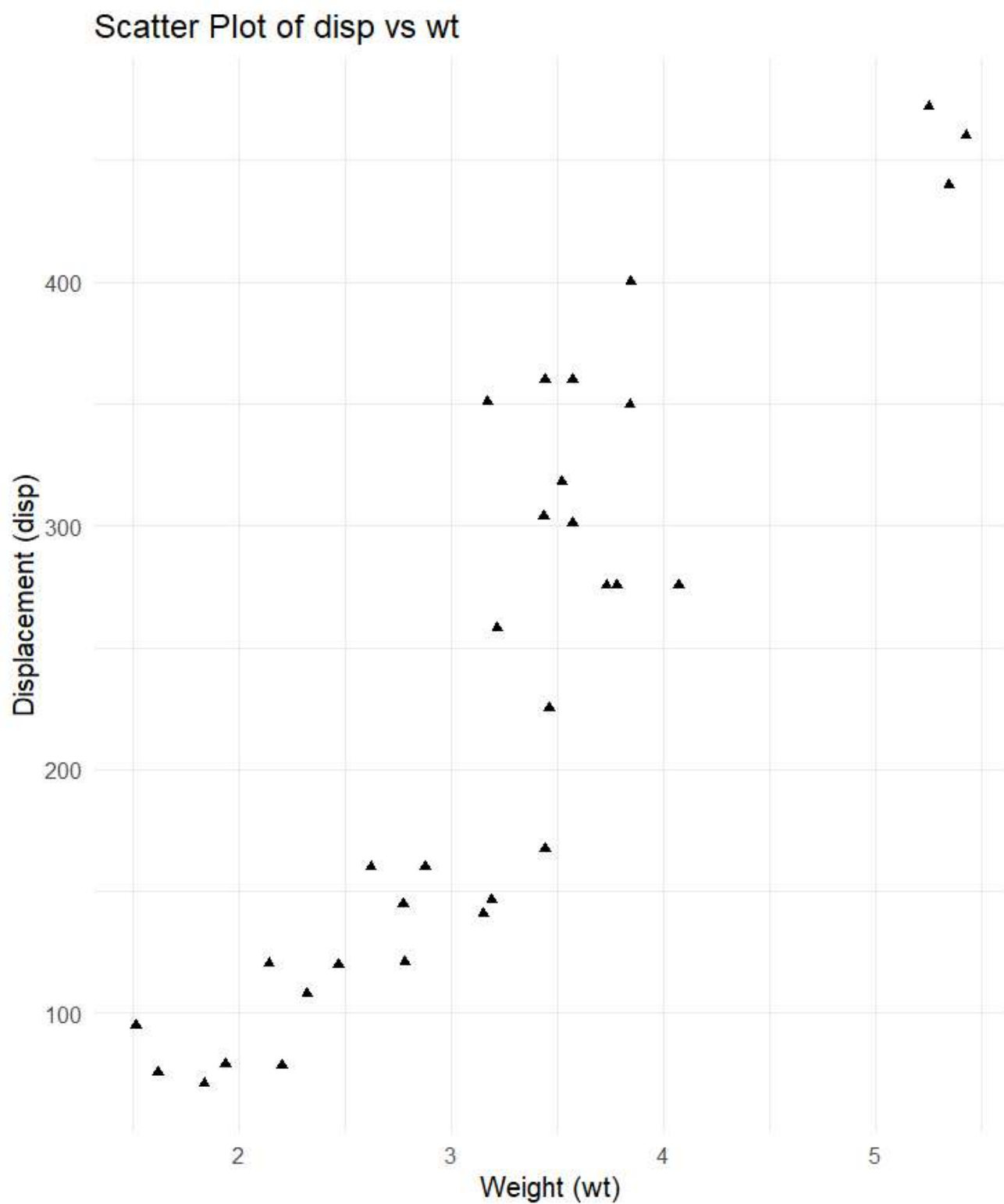


改变散点图中点的设置

- 改变点的形状
 - 代码如下：

```
ggplot(mtcars, aes(x = wt, y = disp)) +  
  geom_point(shape = 17) + # 修改点形状  
  labs(  
    x = "Weight (wt)",  
    y = "Displacement (disp)",  
    title = "Scatter Plot of disp vs wt"  
  ) +  
  theme_minimal()
```

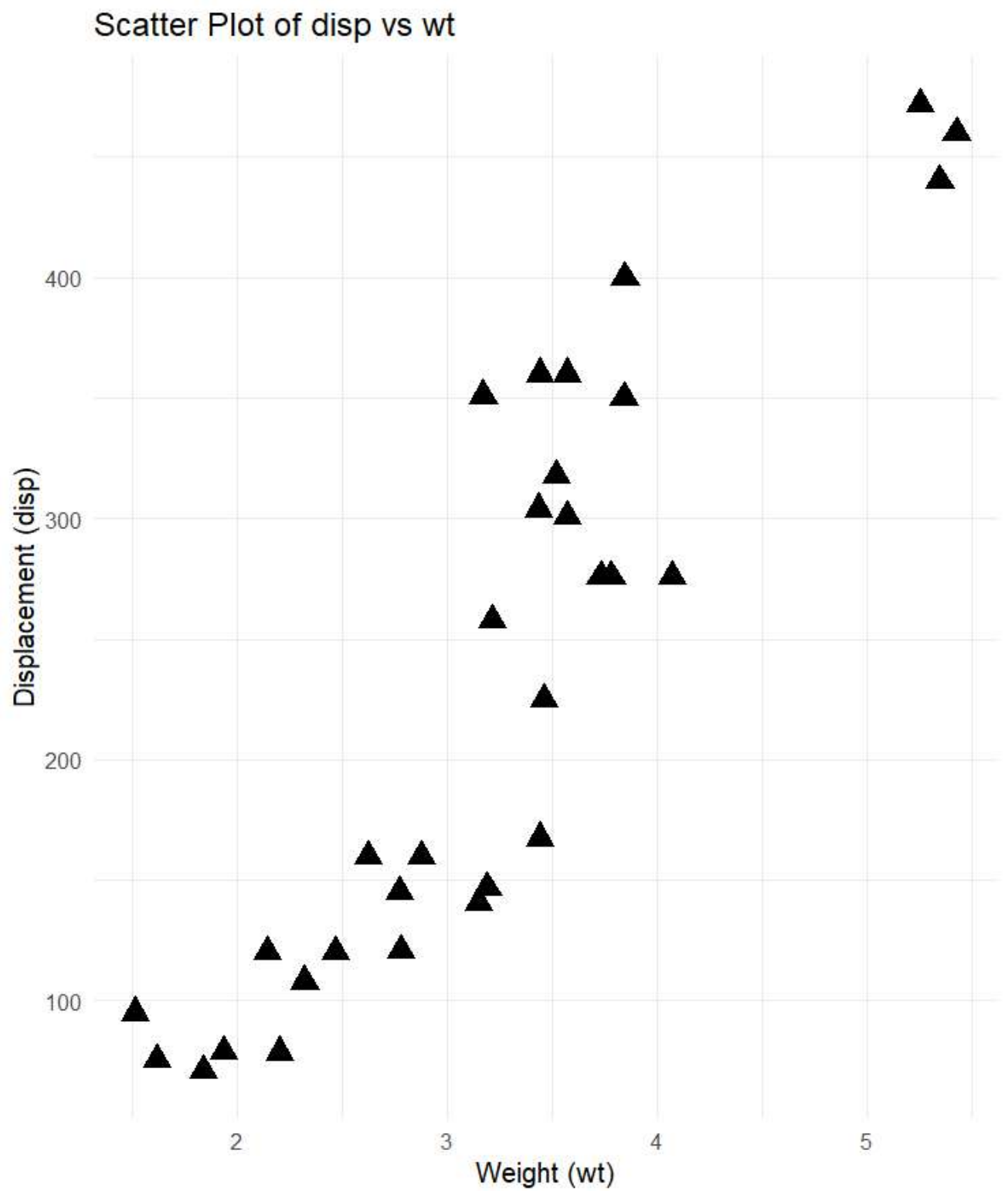
。绘制结果如下：



- 改变点的尺寸
 - 代码如下：

```
ggplot(mtcars, aes(x = wt, y = disp)) +  
geom_point(shape = 17, size = 4) + # 调整点形状 + 放大尺寸  
labs(  
  x = "Weight (wt)",  
  y = "Displacement (disp)",  
  title = "Scatter Plot of disp vs wt"  
) +  
theme_minimal()
```

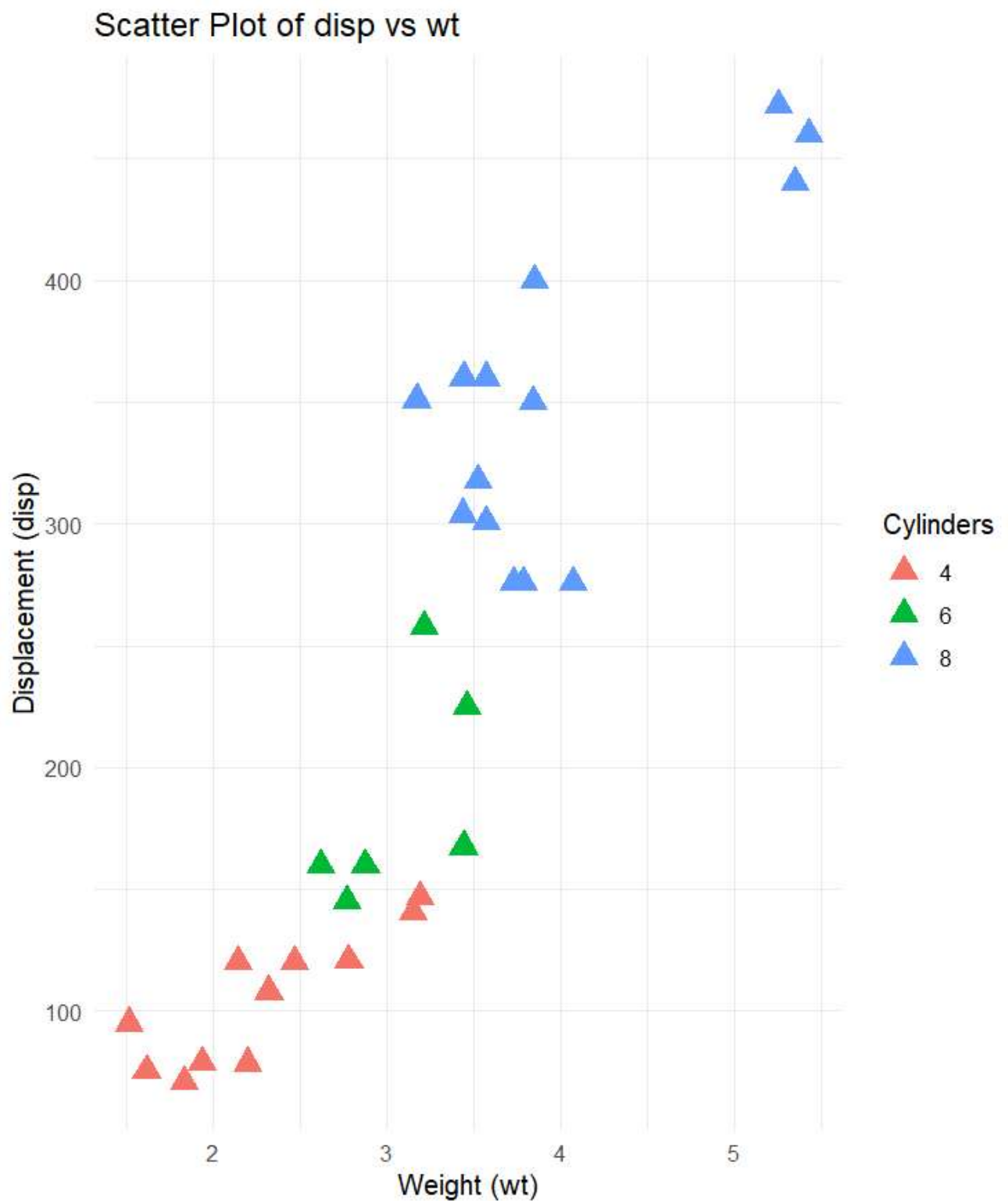
- 绘制结果如下：



- 改变点的颜色
 - 代码如下：

```
ggplot(mtcars, aes(x = wt, y = disp, color = factor(cyl))) +  
  geom_point(shape = 17, size = 4) +  
  labs(  
    x = "Weight (wt)",  
    y = "Displacement (disp)",  
    color = "Cylinders",  
    title = "Scatter Plot of disp vs wt"  
  ) +  
  theme_minimal()
```

。 绘制结果如下：

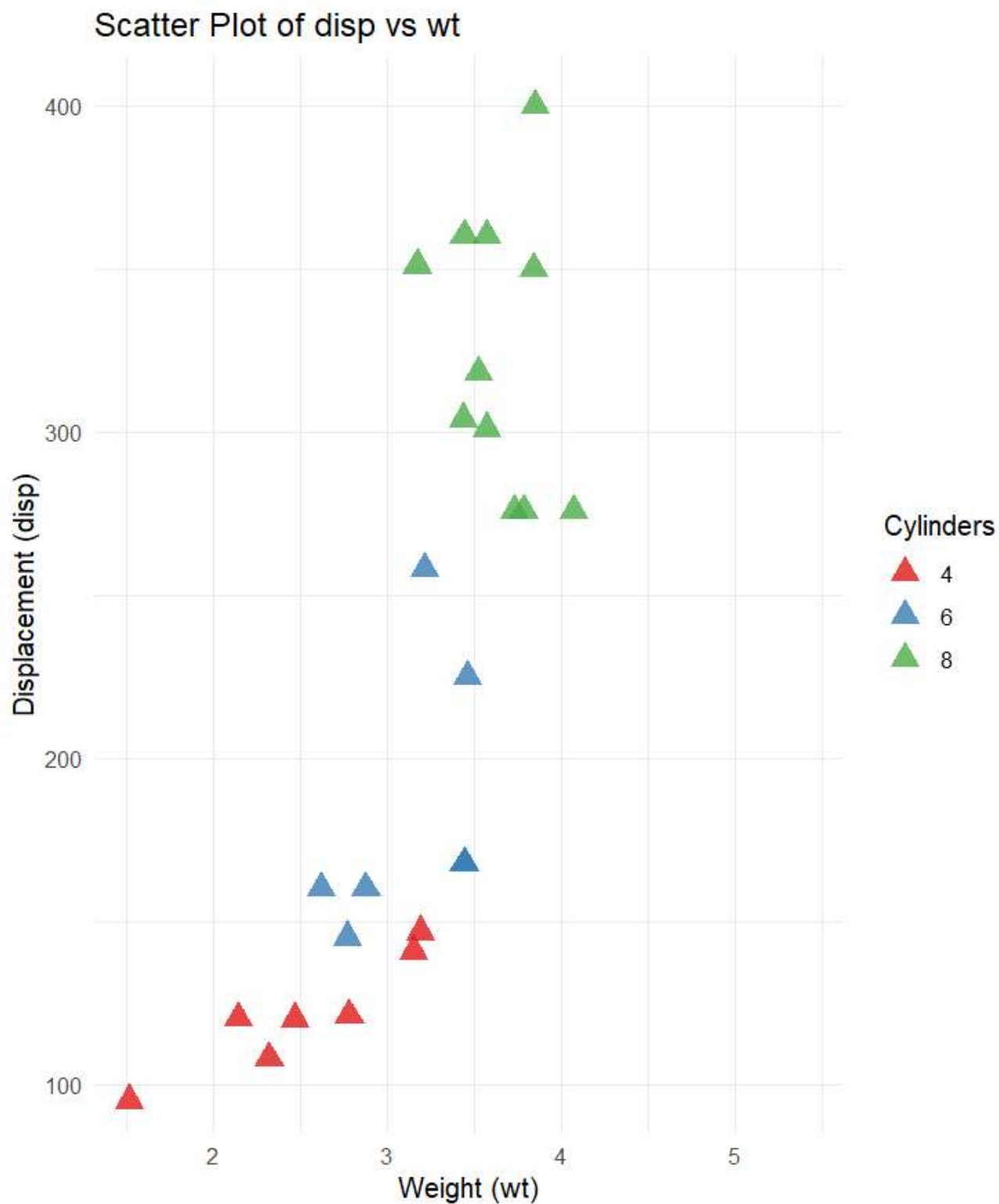


数值范围与文本显示

- 在图像中指定显示的数值范围
 - 代码如下：

```
ggplot(mtcars, aes(x = wt, y = disp, color = factor(cyl))) +  
geom_point(shape = 17, size = 4, alpha = 0.8) +  
coord_cartesian(ylim = c(100, 400)) +  
scale_color_brewer(palette = "Set1") +  
labs(  
  x = "Weight (wt)",  
  y = "Displacement (disp)",  
  color = "Cylinders",  
  title = "Scatter Plot of disp vs wt"  
) +  
theme_minimal()
```

- 绘制结果如下：



- 在图像中增加标题 - 略（上述图像绘制时已经实现了）

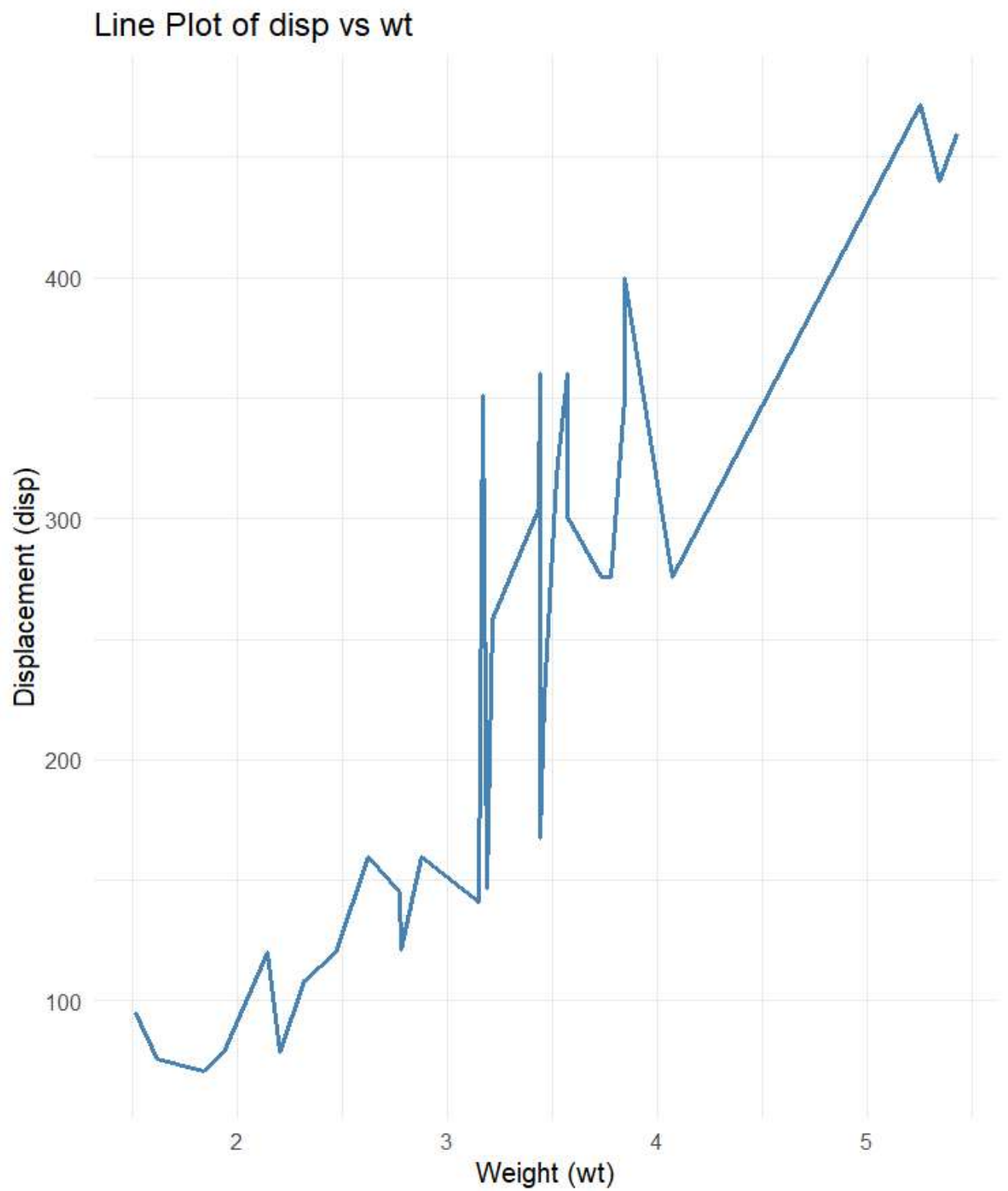
绘制其他类型的图像

绘制折线图

- 基础折线图
 - 代码如下：

```
mtcars %>%  
  arrange(wt) %>% # 按 wt 排序  
  ggplot(aes(x = wt, y = disp)) +  
  geom_line(color = "steelblue", linewidth = 1) +  
  labs(  
    x = "Weight (wt)",  
    y = "Displacement (disp)",  
    title = "Line Plot of disp vs wt"  
  ) +  
  theme_minimal()
```

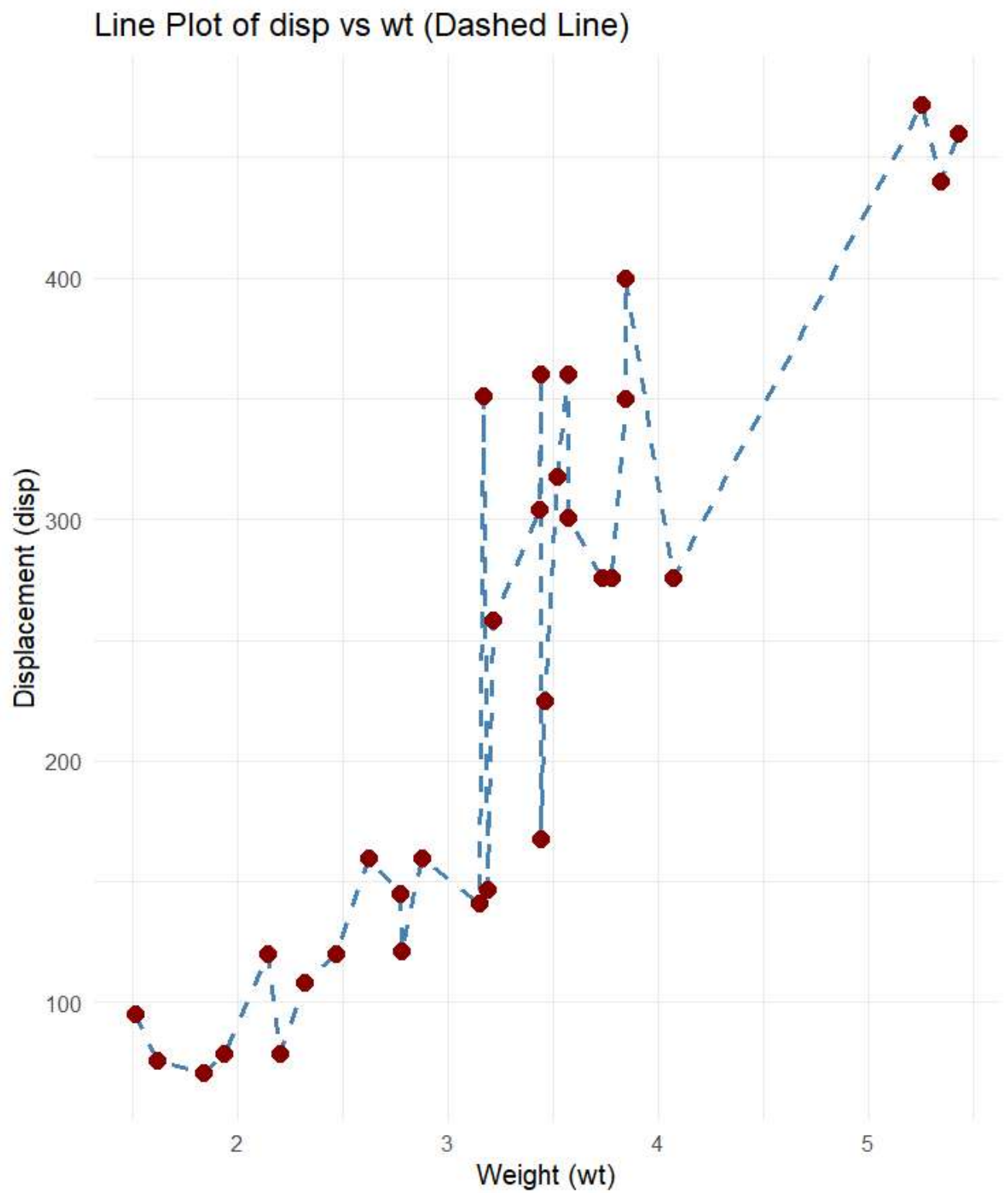
- 绘制结果如下：



- 改变线的类型
 - 代码如下：

```
mtcars %>%  
  arrange(wt) %>%  
  ggplot(aes(x = wt, y = disp)) +  
  geom_line(color = "steelblue", linewidth = 1, linetype = "dashed") +  
  geom_point(size = 3, color = "darkred") +  
  labs(  
    x = "Weight (wt)",  
    y = "Displacement (disp)",  
    title = "Line Plot of disp vs wt (Dashed Line)"  
  ) +  
  theme_minimal()
```

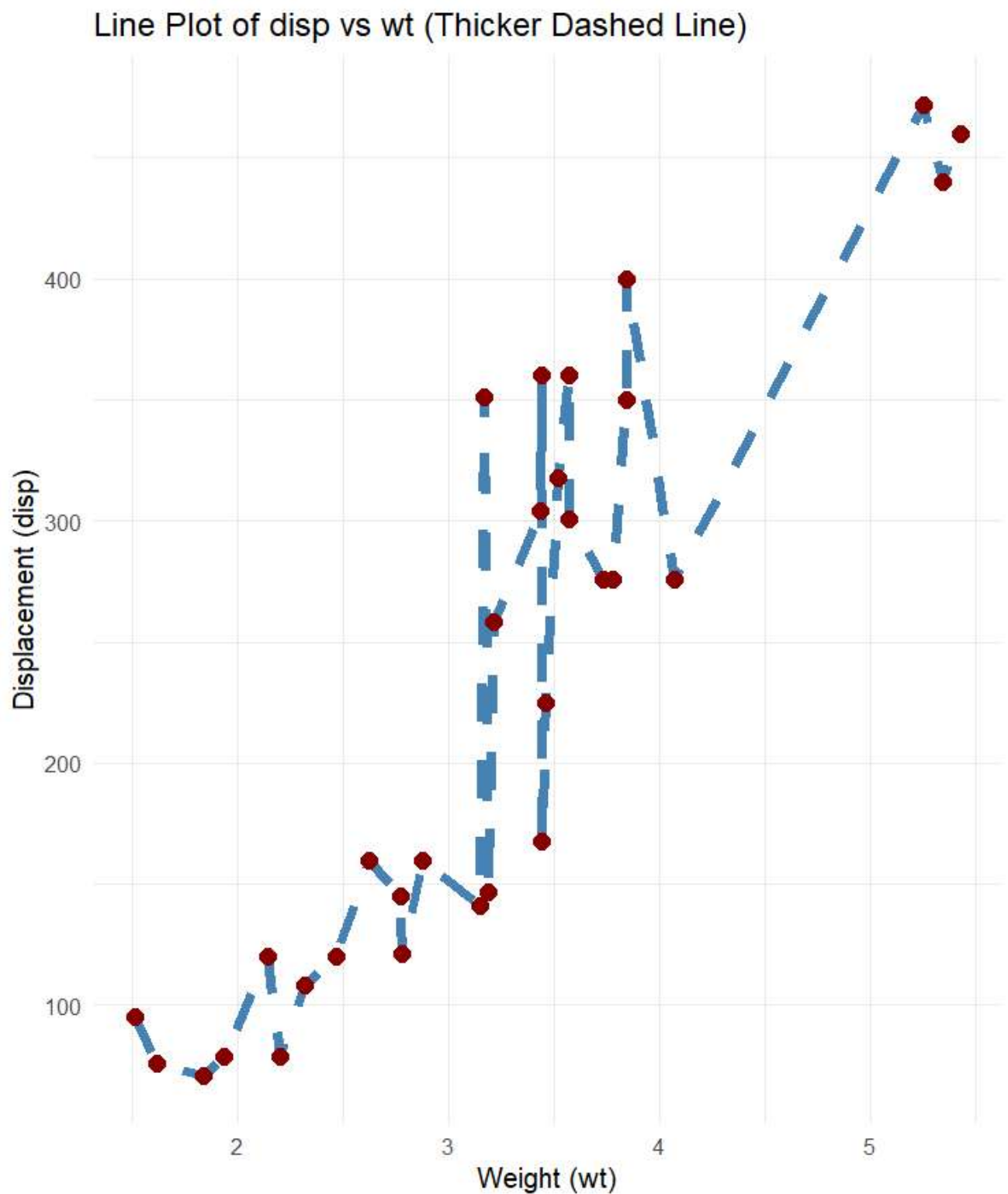
- 绘制结果如下：



- 改变线的粗细
 - 代码如下：

```
mtcars %>%  
  arrange(wt) %>%  
  ggplot(aes(x = wt, y = disp)) +  
  geom_line(  
    color = "steelblue",  
    linetype = "dashed",  
    linewidth = 2 # 加粗线条  
  ) +  
  geom_point(size = 3, color = "darkred") +  
  labs(  
    x = "Weight (wt)",  
    y = "Displacement (disp)",  
    title = "Line Plot of disp vs wt (Thicker Dashed Line)"  
  ) +  
  theme_minimal()
```

- 绘制结果如下：

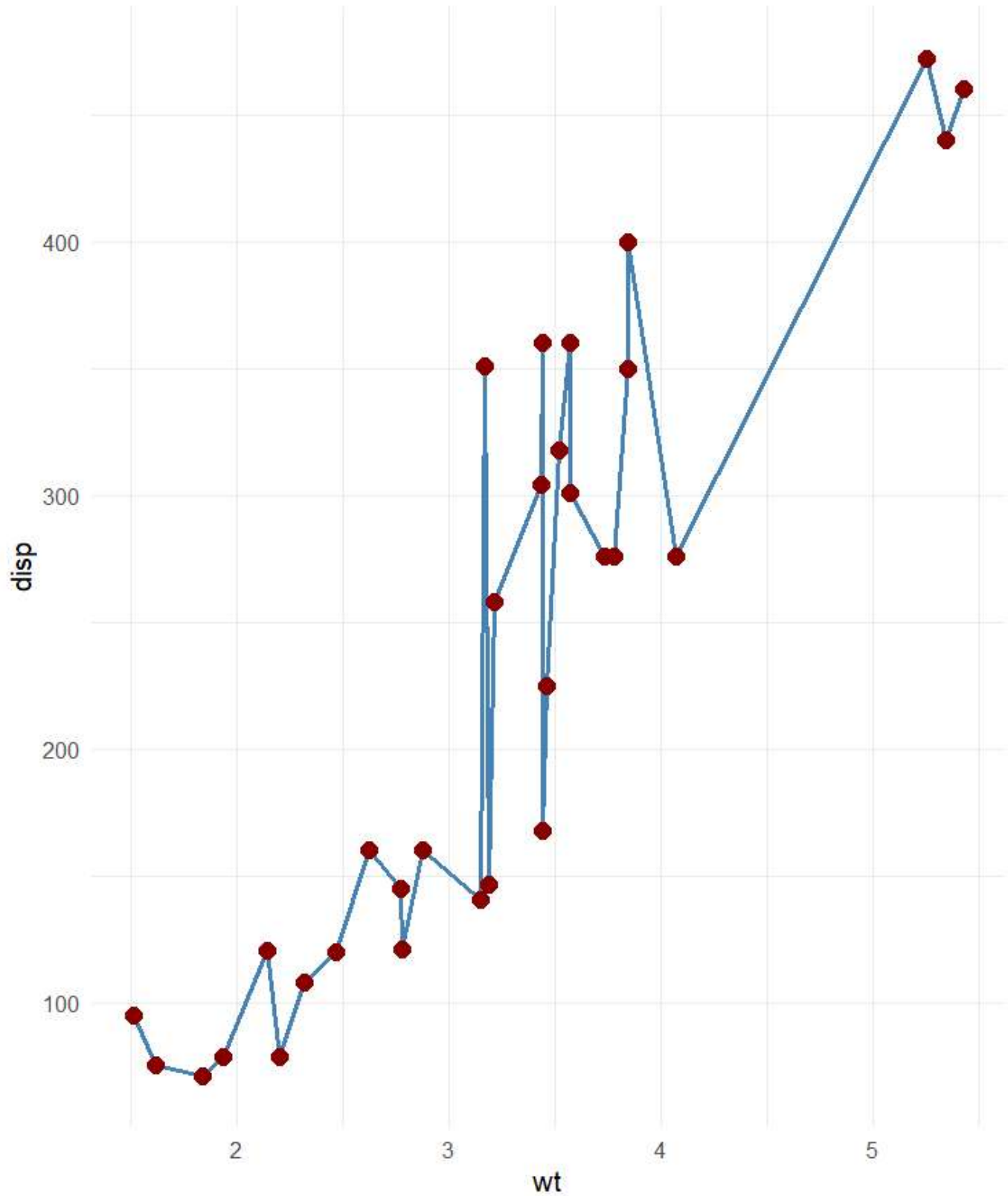


绘制点线图

- 代码如下：

```
mtcars %>%  
  arrange(wt) %>%  
  ggplot(aes(x = wt, y = disp)) +  
  geom_line(color = "steelblue", linewidth = 1) +  
  geom_point(size = 3, color = "darkred") +  
  theme_minimal()
```

- 绘制结果如下：



批量绘图

- 代码如下：

```
types <- c("p", "l", "b", "c", "o", "h", "s", "n")

df <- map_dfr(types, ~ mtcars %>%
  arrange(wt) %>%
  mutate(type = .x))

ggplot(df, aes(x = wt, y = disp)) +

# p: points
geom_point(data = ~ filter(.x, type == "p")) +

# l: line
geom_line(data = ~ filter(.x, type == "l")) +

# b: both (points + line)
geom_line(data = ~ filter(.x, type == "b")) +
geom_point(data = ~ filter(.x, type == "b")) +

# c: line without points (近似)
geom_line(data = ~ filter(.x, type == "c"), linetype = "dashed") +

# o: overplotted (line + point)
geom_line(data = ~ filter(.x, type == "o")) +
geom_point(data = ~ filter(.x, type == "o")) +

# h: histogram-like vertical lines
geom_segment(
  data = ~ filter(.x, type == "h"),
  aes(xend = wt, y = 0, yend = disp)
) +

# s: step
geom_step(data = ~ filter(.x, type == "s")) +

# n: nothing (只保留坐标轴)
geom_blank(data = ~ filter(.x, type == "n")) +

facet_wrap(~ type, ncol = 3) +
labs(x = "mtcars$wt", y = "mtcars$disp") +
theme_minimal()
```

- 绘制结果如下：

